

HUNDRED DAYS OF CODE

DAY 8

Question 1:

Write a C program to find the number of spherical balls that can be made from a larger sphere when both radii are given, assuming exact volume division.

STEP 1 – REQUIRED INPUT, OUTPUT, and FORMULA :

We are given radius of a small and a large sphere with their respective volumes . We need to find the number of spherical balls that can be made out of the large sphere of a given volume.

INPUT :

Let the radius of larger sphere be : R

Let the radius of each small ball be : r

OUTPUT:

Number of spherical balls

Formula : $V = \frac{4\pi r^3}{3}$; where V is volume of larger sphere.

$$\text{Number of balls} = (R / r)^3$$

STEP 2 – TEST CASE TABLE

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	R = 6, r = 2	27 balls	27 balls	PASS
2	R = 10, r = 5	8 balls	8 balls	PASS
3	R = 7.5, r = 3.5	10 balls	10 balls	PASS

STEP 3 – ALGORITHM

1. Start.
2. Read the radius R of the larger sphere.
3. Read the radius r of each small ball.
4. Check that both radii are positive and $R \geq r$.
5. Calculate Number of balls = $(R / r) \times (R / r) \times (R / r)$.
6. Display the number of spherical balls.
7. Stop.

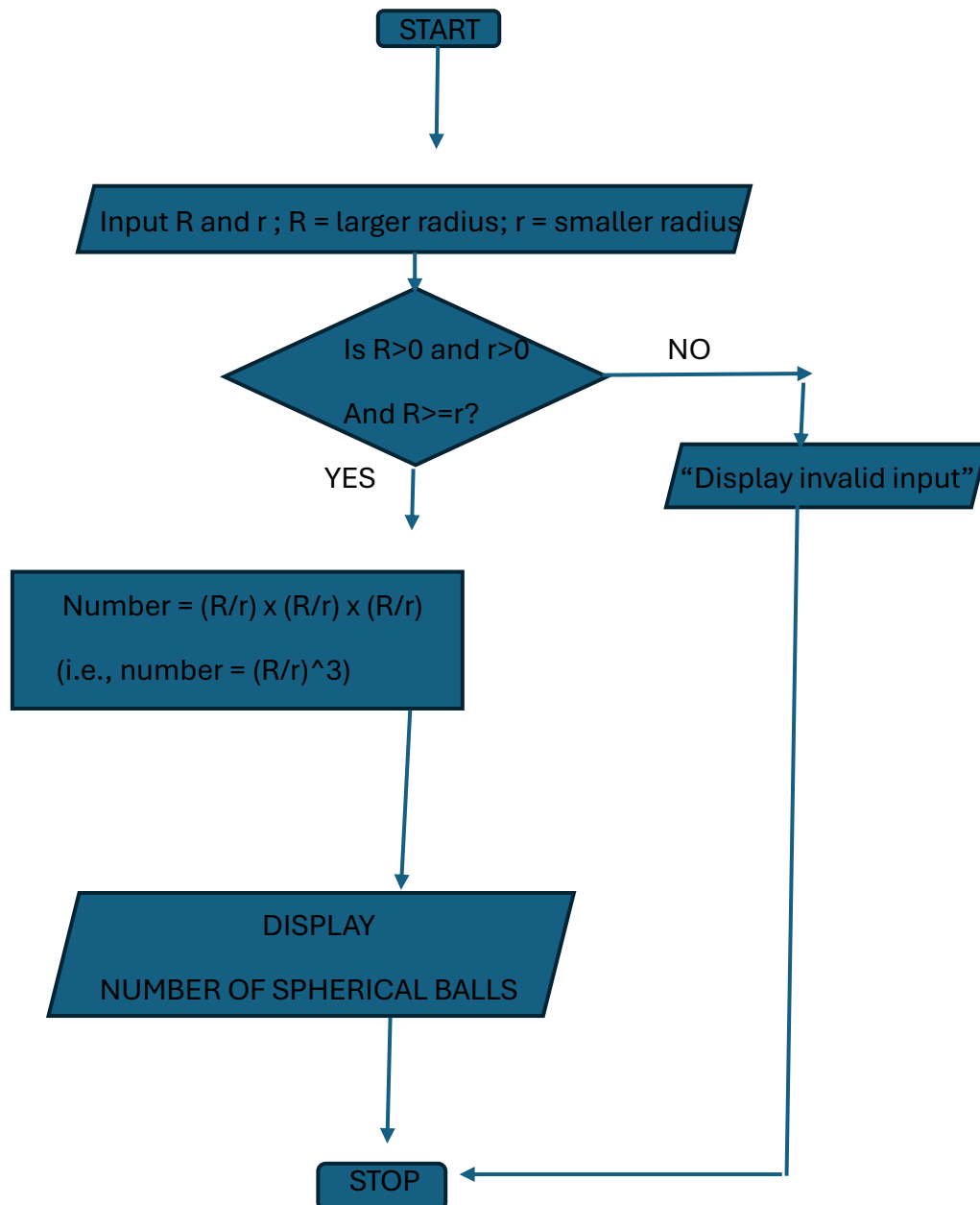
Algorithm evaluation:

For all three test cases, the algorithm produces the expected results: **27, 8, and 27**. Therefore, the algorithm passes all test cases.

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STEP 4- FLOWCHART



Flowchart evaluation:

Test Case	Flowchart Result
1: R = 6, r = 2	27 — Pass
2: R = 10, r = 5	8 — Pass
3: R = 7.5, r = 3.5	10 — Pass

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STEP 5 – REQUIRED C PROGRAM

```
#include <stdio.h>

int main(void) {
    double R, r;

    printf("Enter radius of larger sphere: ");
    scanf("%lf", &R);

    printf("Enter radius of each small ball: ");
    scanf("%lf", &r);

    if (R <= 0 || r <= 0 || R < r) {
        printf("Invalid input. Radii must be positive and R >= r.\n");
        return 1;
    }

    double balls = (R / r) * (R / r) * (R / r);

    printf("Number of spherical balls = %.0f\n", balls);

    return 0;
}
```

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STEP 6 – COMPILATION , EXECUTION

```
(kali㉿kali)-[~]  
$ gcc q81.c -o q81  
  
(kali㉿kali)-[~]  
$ ./q81  
Enter radius of larger sphere: 6  
Enter radius of each small ball: 2  
Number of spherical balls = 27  
  
(kali㉿kali)-[~]  
$ ./q81  
Enter radius of larger sphere: 10  
Enter radius of each small ball: 5  
Number of spherical balls = 8  
  
(kali㉿kali)-[~]  
$ ./q81  
Enter radius of larger sphere: 7.5  
Enter radius of each small ball: 3.5  
Number of spherical balls = 10  
  
(kali㉿kali)-[~]  
$
```

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QUESTION 2 - Write a C program to find the length of a common chord when radius and distance from center are given.

STEP 1 - Input:

- Radius, r
- Distance from center to chord, d

Output:

- Length of the chord

Formula:

For half of the chord, a right triangle is formed:

$$\text{Half chord} = \sqrt{r^2 - d^2}$$

Therefore:

$$\text{Chord length} = 2\sqrt{r^2 - d^2}$$

The valid input condition is $r > 0$ and $0 \leq d \leq r$.

STEP 2 – TEST CASE TABLE

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	r= 5, d = 3	8.00	8.00	PASS
2	r = 10, d = 6	16.00	16.00	PASS
3	r = 8, d = 0	16.00	16.00	PASS

STEP 3 – ALGORITHM

1. Start.
2. Read the radius r.
3. Read the distance d from the center to the chord.
4. Check that $r > 0$ and $0 \leq d \leq r$.
5. Calculate $\text{length} = 2 \times \sqrt{r^2 - d^2}$.
6. Display the length of the chord.

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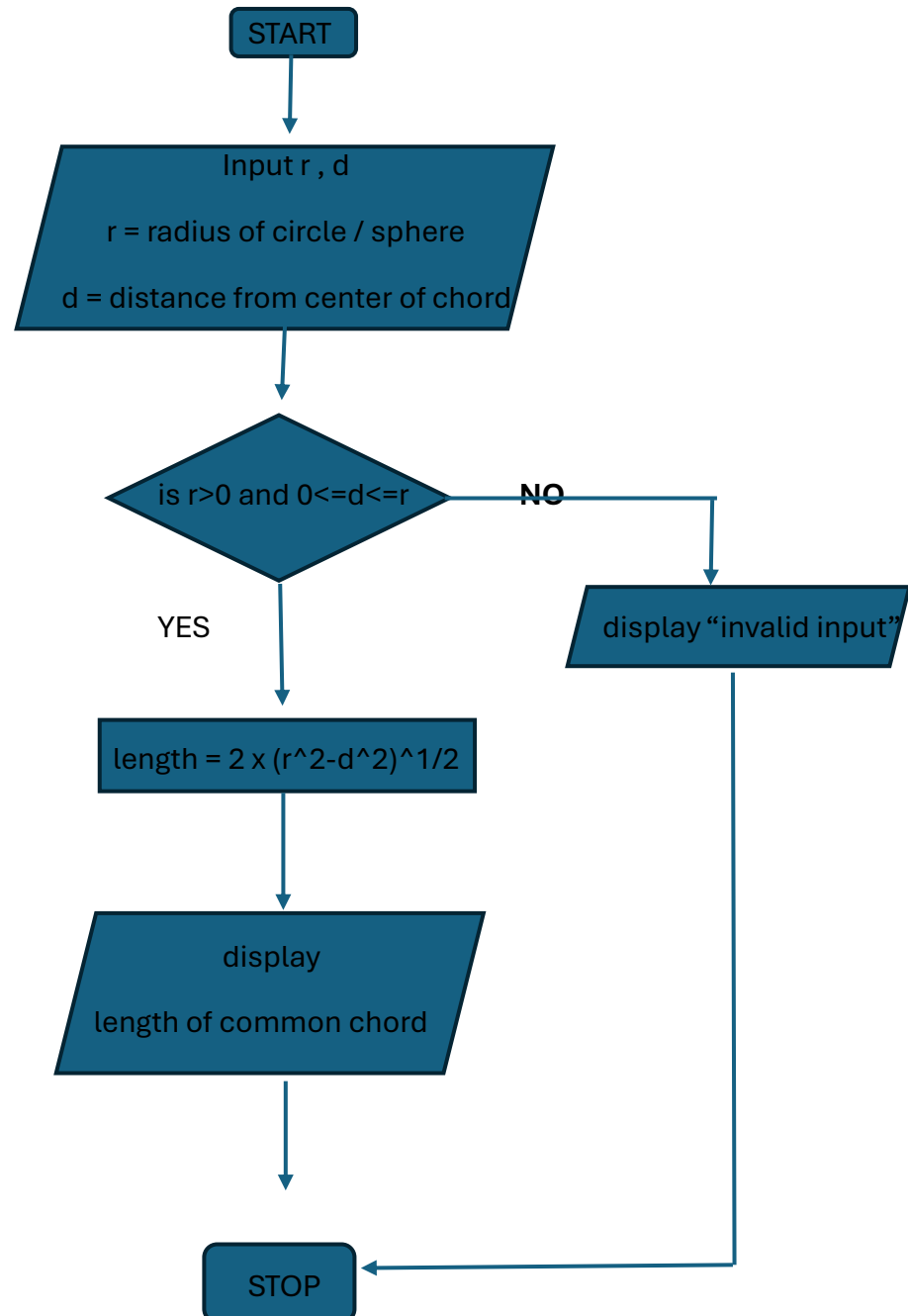
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7. Stop.

Algorithm evaluation:

The algorithm gives **8.00**, **16.00**, and **16.00** for the three test cases, matching the expected outputs. Therefore, it passes all test cases.

STEP 4 – FLOWCHART



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Flowchart evaluation:

Test Case	Flowchart Result
-----------	------------------

1: r = 5, d = 3	8.00 — Pass
-----------------	-------------

2: r = 10, d = 6	16.00 — Pass
------------------	--------------

3: r = 8, d = 0	16.00 — Pass
-----------------	--------------

STEP 5 – REQUIRED C PROGRAM

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main(void) {
```

```
    double r, d
```

```
    printf("Enter radius of sphere: ");
```

```
    scanf("%lf", &r)
```

```
    printf("Enter distance from center to chord: ");
```

```
    scanf("%lf", &d);
```

```
    if (r <= 0 || d < 0 || d > r) {
```

```
        printf("Invalid input. Use r > 0 and 0 <= d <= r.\n");
```

```
        return 1;
```

```
    }
```

```
    double length = 2 * sqrt(r * r - d * d);
```

```
    printf("Length of common chord = %.2f\n", length);
```

```
    return 0;
```

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}

STEP 6 – COMPILATION AND EXECUTION

```
(kali㉿kali)-[~]  
$ gcc q82.c -o q82 -lm  
  
(kali㉿kali)-[~]  
$ ./q82  
Enter radius of sphere: 5  
Enter distance from center to chord: 3  
Length of common chord = 8.00  
  
(kali㉿kali)-[~]  
$ ./q82  
Enter radius of sphere: 10  
Enter distance from center to chord: 6  
Length of common chord = 16.00  
  
(kali㉿kali)-[~]  
$ ./q82  
Enter radius of sphere: 8  
Enter distance from center to chord: 0  
Length of common chord = 16.00  
  
(kali㉿kali)-[~]  
$
```

THANKYOU